Haokai Ding

Undergraduate Researcher · Robotics

Shenzhen, China

ditang0125@gmail.com | haokaiding.github.io | HaokaiDing | ORCID: 0009-0001-6158-9897

Education

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) — Abu Dhabi, UAE
M.Sc. in Robotics (Incoming) — Fall 2026

Tsinghua University | Shenzhen X-Institute — Shenzhen, China
Tsien Excellence in Engineering Program (TEEP) – X Scholar — Sep. 2023 – Present

- Joint training program in underactuated robotics and mechanism design.

Shenzhen Technology University – Future Technology School — Shenzhen, China
B.Eng. in Electronic Science and Technology — Sep. 2022 – Jul. 2026

- GPA: 84.6/100. Research focus: LIBS spectroscopy and robotics systems.

Research Experience

State Key Lab of Mechanical Systems & Vibration, Shanghai Jiao Tong University — Shanghai, China
Visiting Student – UAV Perching and Aerial Manipulation — Feb. 2026 – Present

- Advisor: Prof. Wei Dong.
- Working on UAV perching and contact-rich aerial manipulation.

Shenzhen X-Institute, Tsinghua University — Shenzhen, China
Undergraduate Researcher – Underactuated Robotic Grippers — 2023 – Present

- Designed and tested underactuated grippers based on semi-Peaucellier linkages.
- Developed idle-stroke and delay-triggering mechanisms; validated prototypes via simulation and physical experiments.

Shenzhen Technology University — Shenzhen, China
Undergraduate Researcher – LIBS Spectroscopy for Heavy-Metal Analysis — 2022 – 2023

- Built and optimized a LIBS system for rapid heavy-metal adsorption analysis.
- Improved the calibration pipeline and acquisition accuracy.

Publications

† Co-first author * Corresponding author Listed in reverse chronological order by IEEE Xplore date added.

Selected Conference Papers (Robotics)

Y. Chen, H. Luan, **H. Ding**, W. Zhang* — IEEE RAAI 2025, Singapore
[P1] An Idle-Stroke Underactuated Gripper with Independent Double-Parallelogram Linkages for Pinching and Adaptive Grasping — Mar. 2026

H. Ding†, X. Zhong, W. Huang, T. Yang*, W. Zhang* — IEEE ROBIO 2025, Bangkok, Thailand
[P2] BioCrest – A Flexible Adaptive Robotic Hand Based on Idle-Stroke Mechanism — Feb. 2026

H. Ding, H. Luan, T. Yang*, W. Zhang* — IEEE ICCMA 2025, Macau, China
[P3] GLINT: An Idle-Stroke Grasp-and-Lift Hand for In-Hand Manipulation — Feb. 2026

H. Ding†, J. Fan, Z. Zhu, Y. Zhao, K. Chen*, W. Zhang* — IEEE ICARM 2025, Hangzhou, China
[P4] Peaucellier Gripper: A Novel Underactuated Gripper for Linear Pinching and Self-Adaptive Grasp — Dec. 2025

H. Ding†, Y. Chen†, D. Jia, W. Zhang* — IEEE ICIA 2025, Wuhan, China
[P5] An Underactuated Gripper with Grasshopper-Inspired Linkages and Delay Triggering for Pinching and Adaptive Grasping — Dec. 2025

H. Ding†, W. Zhang* — IEEE/RSJ IROS 2025, Hangzhou, China
[P6] A Novel Gripper with Semi-Peaucellier Linkage and Idle-Stroke Mechanism for Linear Pinching and Self-Adaptive Grasping — Nov. 2025

H. Ding†, Z. Chen, T. Yang*, W. Zhang* IEEE CASE 2025, Los Angeles, USA

[P7] Semi-Peaucellier Linkage and Differential Mechanism for Linear Pinching and Self-Adaptive Grasping Sep. 2025

Other Publications

Y. Liu†, H. Ding†, H. Chen, H. Gao, J. Yu, F. Mo, N. Wang* Polymers (MDPI) 17(15):2136

[P8] Recent Progress on the Research of 3D Printing in Aqueous Zinc-Ion Batteries Aug. 2025

H. Xie, L. You, X. Fang, L. Li, H. Ding, Z. Zhou, G. Zhang, D. Zhang E3S Web of Conferences, 615 (2025) 02010

[P9] Rapid Analysis of Heavy-Metal Element Adsorption by SCG Based on LIBS Technology Feb. 2025

Honors & Awards

International Awards

- 2025 Undergraduate Champion, ASME Student Mechanism & Robot Design Competition — USA

National & Domestic Awards

- 2025 National Scholarship, Ministry of Education of China — China
- 2024 President's Award, Shenzhen Technology University — Shenzhen, China
- 2024 Research and Innovation Award (First Prize), Shenzhen Technology University — Shenzhen, China
- 2023 Gold Prize, 9th China "Internet+" Innovation Competition (Guangdong Division) — Guangdong, China
- 2023 Second Prize, 17th "Challenge Cup" Extracurricular Science Competition (Guangdong Division) — Guangdong, China
- 2023 Research and Innovation Award (First Prize), Shenzhen Technology University — Shenzhen, China

Professional Service

- 2025 Conference Reviewer, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- 2025 Session Chair, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) — Hangzhou, China
- 2025 Conference Reviewer, IEEE International Conference on Automation Science and Engineering (CASE)